

$$\begin{array}{c}
\begin{array}{cc}
\mathcal{K}_{0^+}^{\#1} & \mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{0^+}^{\#1}{}^\dagger & \begin{array}{cc} \frac{-3Y_1 k^2}{4} & 0 \end{array} \\
\mathcal{K}_{0^+}^{\#1}{}^{\alpha\beta\chi} & \begin{array}{cc} 0 & 0 \end{array}
\end{array}
\begin{array}{cccc}
\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta} & \mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta} & \mathcal{K}_{1^+}^{\#1}{}_{\alpha} & \mathcal{K}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta} & \begin{array}{cc} \frac{Y_2 k^2}{2} & \frac{Y_2 k^2}{2\sqrt{2}} \end{array} & \begin{array}{cc} 0 & 0 \end{array} \\
\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta} & \begin{array}{cc} \frac{Y_2 k^2}{2\sqrt{2}} & \frac{Y_2 k^2}{4} \end{array} & \begin{array}{cc} 0 & 0 \end{array} \\
\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha} & \begin{array}{cc} 0 & 0 \end{array} & \begin{array}{cc} 0 & 0 \end{array} \\
\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha} & \begin{array}{cc} 0 & 0 \end{array} & \begin{array}{cc} 0 & 0 \end{array}
\end{array}
\begin{array}{cc}
\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} & \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta} & \begin{array}{cc} \frac{3k^2{}^{(4)}\kappa_{15}}{2} & 0 \end{array} \\
\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi} & \begin{array}{cc} 0 & \frac{3k^2{}^{(4)}\kappa_{15}}{2} \end{array}
\end{array}
\end{array}$$

$$\begin{array}{c}
\begin{array}{cc}
\mathcal{J}_{0^+}^{\#1} & \mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{0^+}^{\#1}{}^\dagger & \begin{array}{cc} \frac{1}{\text{Det}(0^+)} & 0 \end{array} \\
\mathcal{J}_{0^+}^{\#1}{}^{\alpha\beta\chi} & \begin{array}{cc} 0 & 0 \end{array}
\end{array}
\begin{array}{cccc}
\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta} & \mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta} & \mathcal{J}_{1^+}^{\#1}{}_{\alpha} & \mathcal{J}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta} & \begin{array}{cc} \frac{2}{3\text{Det}(1^+)} & \frac{\sqrt{2}}{3\text{Det}(1^+)} \end{array} & \begin{array}{cc} 0 & 0 \end{array} \\
\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta} & \begin{array}{cc} \frac{\sqrt{2}}{3\text{Det}(1^+)} & \frac{1}{3\text{Det}(1^+)} \end{array} & \begin{array}{cc} 0 & 0 \end{array} \\
\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha} & \begin{array}{cc} 0 & 0 \end{array} & \begin{array}{cc} 0 & 0 \end{array} \\
\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha} & \begin{array}{cc} 0 & 0 \end{array} & \begin{array}{cc} 0 & 0 \end{array}
\end{array}
\begin{array}{cc}
\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta} & \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta} & \begin{array}{cc} \frac{2}{3k^2{}^{(4)}\kappa_{15}} & 0 \end{array} \\
\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi} & \begin{array}{cc} 0 & \frac{2}{3k^2{}^{(4)}\kappa_{15}} \end{array}
\end{array}
\end{array}$$

Abbreviations used in matrices

$$Y_1 == 2 \begin{pmatrix} 4 \\ \kappa_1 \end{pmatrix} + \begin{pmatrix} 4 \\ \kappa_{15} \end{pmatrix} \ \&\& \ Y_2 == \begin{pmatrix} 4 \\ \kappa_{15} \end{pmatrix} - 2 \begin{pmatrix} 4 \\ \kappa_3 \end{pmatrix} \ \&\& \ \text{Det}(0^+) == -\frac{3}{4} k^2 (2 \begin{pmatrix} 4 \\ \kappa_1 \end{pmatrix} + \begin{pmatrix} 4 \\ \kappa_{15} \end{pmatrix}) \ \&\& \ \text{Det}(1^+) == \frac{3}{4} k^2 (\begin{pmatrix} 4 \\ \kappa_{15} \end{pmatrix} - 2 \begin{pmatrix} 4 \\ \kappa_3 \end{pmatrix})$$

Lagrangian

$$\begin{aligned}
& \begin{pmatrix} 4 \\ \kappa_1 \end{pmatrix} \partial_\beta \mathcal{K}^\delta{}_{\chi\delta} \partial^\chi \mathcal{K}^\alpha{}^\beta + \frac{3}{2} \begin{pmatrix} 4 \\ \kappa_{15} \end{pmatrix} \partial_\chi \mathcal{K}^\delta{}_{\beta\delta} \partial^\chi \mathcal{K}^\alpha{}^\beta + \begin{pmatrix} 4 \\ \kappa_3 \end{pmatrix} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\alpha\chi} + \\
& 3 \begin{pmatrix} 4 \\ \kappa_{15} \end{pmatrix} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi} - 2 \begin{pmatrix} 4 \\ \kappa_3 \end{pmatrix} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi} - \begin{pmatrix} 4 \\ \kappa_3 \end{pmatrix} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\chi\alpha} - 3 \begin{pmatrix} 4 \\ \kappa_{15} \end{pmatrix} \partial^\chi \mathcal{K}^\alpha{}^\beta \partial_\delta \mathcal{K}^\delta{}_{\chi\beta} - \\
& \frac{3}{4} \begin{pmatrix} 4 \\ \kappa_{15} \end{pmatrix} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi} - \frac{1}{2} \begin{pmatrix} 4 \\ \kappa_3 \end{pmatrix} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_{\beta\chi} + \begin{pmatrix} 4 \\ \kappa_{15} \end{pmatrix} \partial_\delta \mathcal{K}^{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \begin{pmatrix} 4 \\ \kappa_{15} \end{pmatrix} \partial_\delta \mathcal{K}^{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$	
$\mathcal{J}_{1^+}^{\#2\alpha} == 0$	3	$\partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi} == 0$	
$\mathcal{J}_{1^+}^{\#1\alpha} == 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta{}_\chi + \partial_\chi \partial^\chi \mathcal{J}^\beta{}_\beta == \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}_{1^+}^{\#1\alpha\beta} == \mathcal{J}_{1^+}^{\#2\alpha\beta}$	3	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} + 2 \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} == 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$	
Total # constraint(s):	10		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{1}{\begin{pmatrix} 4 \\ \kappa_{15} \end{pmatrix} - 2 \begin{pmatrix} 4 \\ \kappa_3 \end{pmatrix}}$
	2	0	$-\frac{1}{\begin{pmatrix} 4 \\ \kappa_{15} \end{pmatrix}}$
	2	0	$\frac{1}{\begin{pmatrix} 4 \\ \kappa_{15} \end{pmatrix}}$
Resolved unitarity condition(s):		(Demonstrably impossible)	